



CarpalSense Instruction Manual

Group 3 Wearable Sensor

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Procedure

1

Place the EMG sensor on your forearm using the band attached to it. Ensure that the cable is facing away from you and that the EMG is placed on the **Flexor Pollicis Longus** muscle

Note: This step may require multiple attempts due to the abundance of muscles in the forearm; refer to Troubleshooting for further assistance

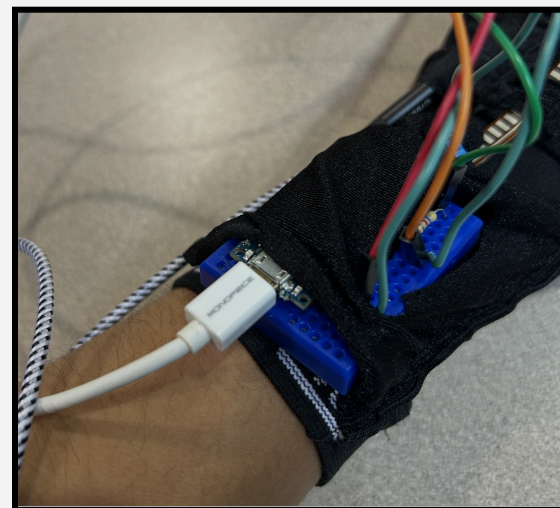
Figure 01: Example placement of EMG and proper positioning on forearm



2

Connect a Micro-B to USB cable to the Arduino Nano 33 IOT (located underneath the housing) but **do not** plug the USB end into the device being connected.

Figure 02: Micro-B Cable being connected to Arduino Nano

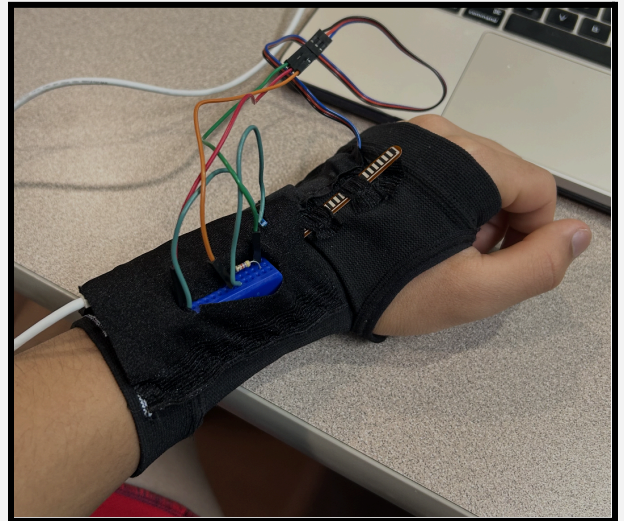


Procedure

3

Gently pull the glove over your left hand. Make sure that the Arduino 33 Nano IOT and the rest of the circuit are facing up when you do so. Double check all sensor positions before beginning any data collection.

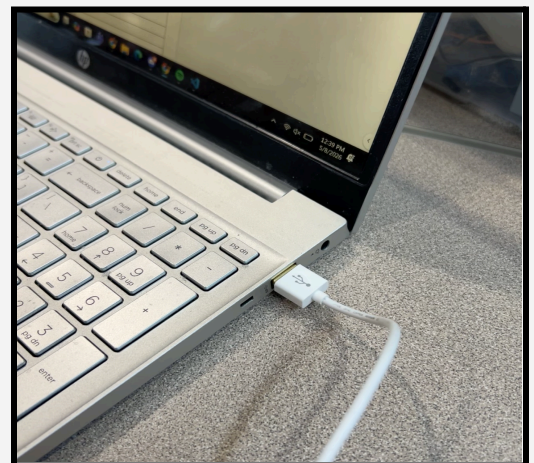
Figure 03: Wearable sensor fully fitted and worn on the hand.



4

Plug in the USB into the computer and place your gloved hand flat on a desk in a flat position for 5-8 seconds. Holding this position will allow the sensor to calibrate the EMG.

Figure 04: Example of USB cable connected to a viable device



Procedure

5

Go to the CarpalSense website and log into your account. If you do not have one, create an account, following the prompts given

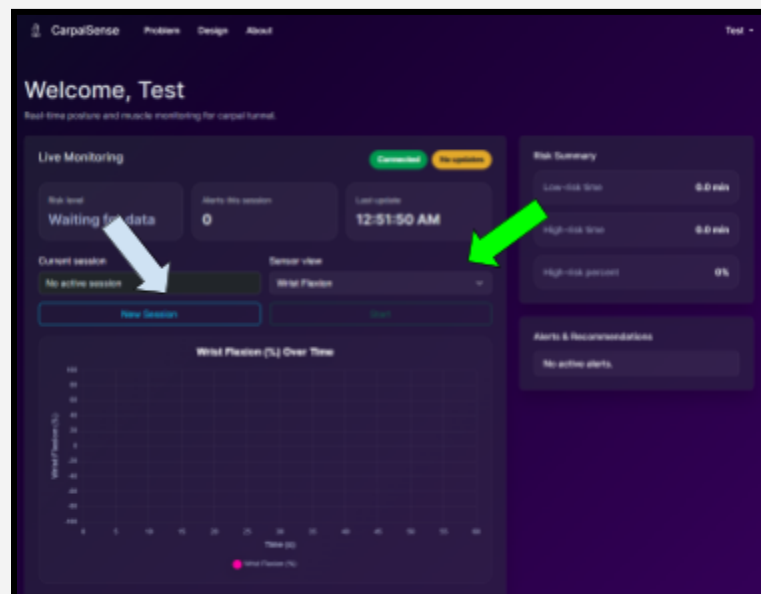
Figure 05: Homescreen of CarpalSense website with the Log-In page highlighted.



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Once logged in, select New Session on the Dashboard and click Start to begin collecting data. To toggle which sensor you see, use the drop-down menu

Figure 06: Example Dashboard with the New Session button (blue arrow) and Sensor drop down menu (green arrow) highlighted



Procedure

7

After the calibration is complete, all FPL and wrist movement will be monitored and shown on the website via graphs (see Data)

Note: Wrist movements involve bending the wrist forward and backwards and the FPL contractions are monitored by clenching the fist and flexing the muscle

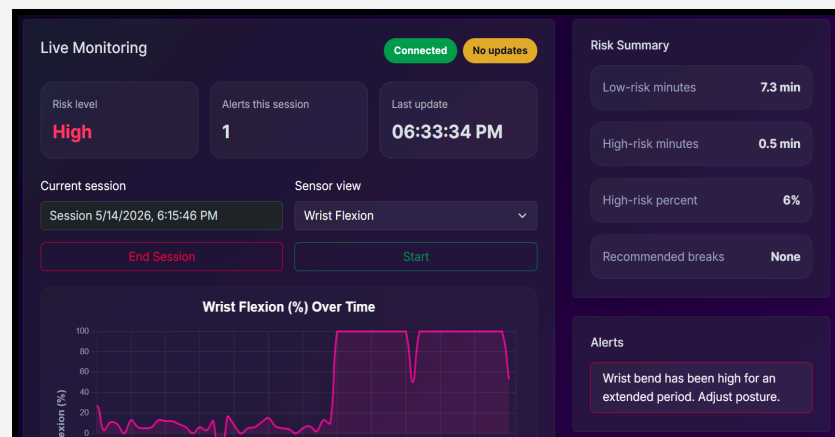
Figure 07: Serial Monitor showing live sensor readings

```
=====EMG SENSOR=====
min: 13300 max: 33518 spread: 152.0% -> Medium
=====FLEX SENSOR=====
557 | 0.45V | 298.5KΩ | -11 | BACKWARD Slight | 1
=====MATH MODEL=====
EMG last 3 sum: 5 passed: YES
Flex last 3 sum: 4 passed: YES
Combined Score: 9 | Counter: 48 | CTS Risk: false
```

8

If the accumulated strain on the wrist reaches a “danger zone”, a notification will be sent out through the website to tell you to take a rest.

Figure 08: Alert on website to show entry into danger zone

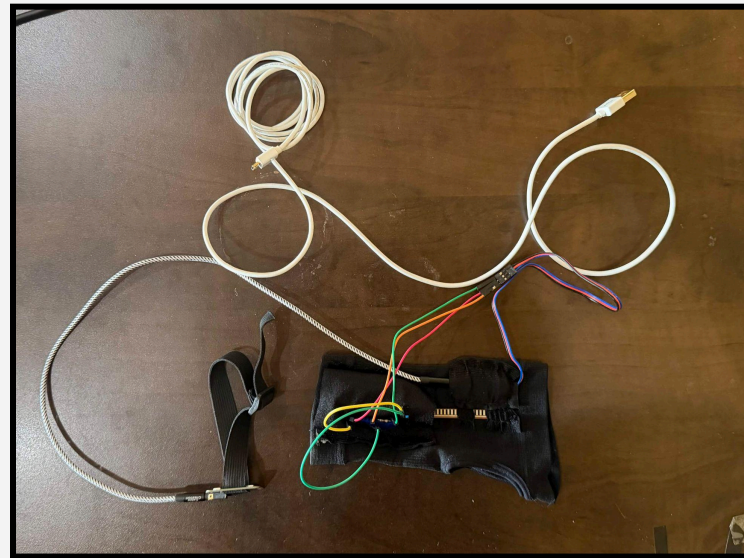


Procedure

9

To remove the sensor, unplug the Micro-B to USB cable from both ports and then gently remove the glove as well as the Dry EMG band from the forearm. Ensure that no cables are displaced in this process

Figure 09: Safely removed sensor as well as other components used (cable and band)



Circuit Designs

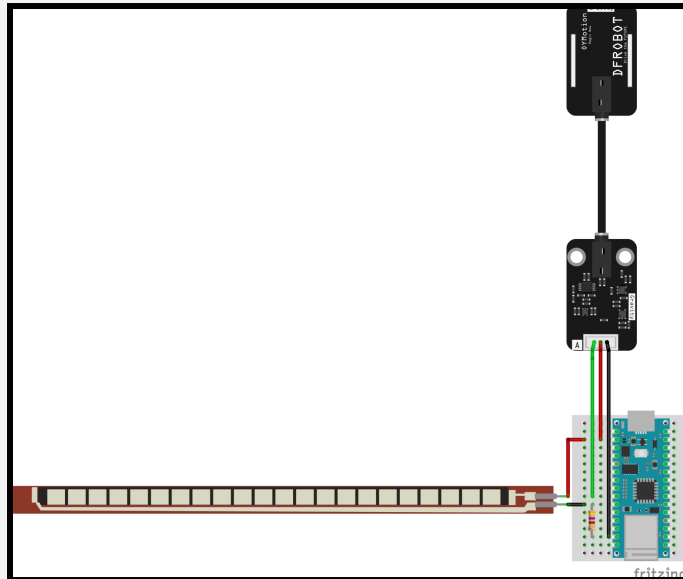


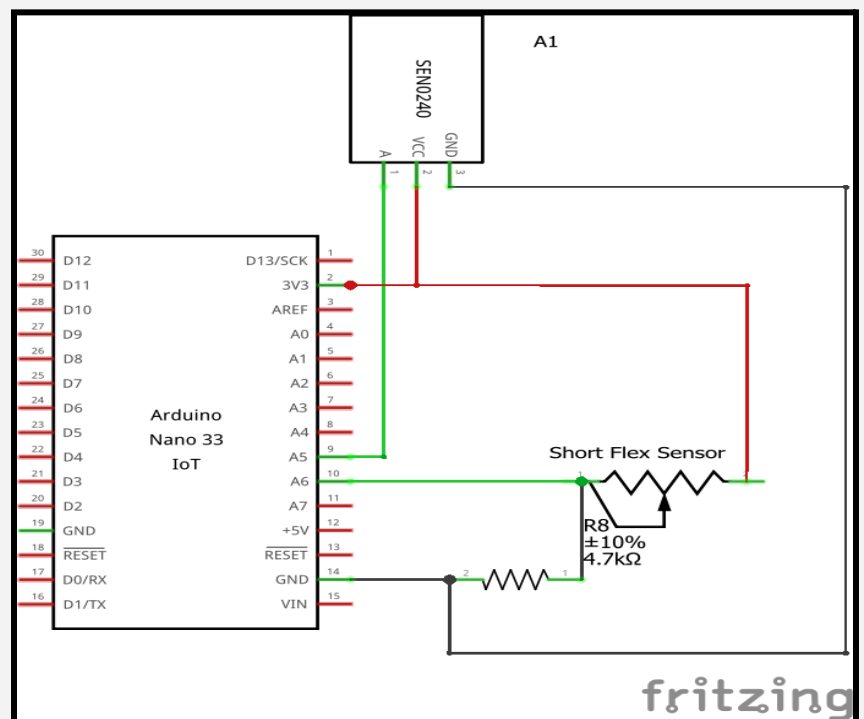
Figure 10: Fritzing Wire Schematic for CarpalSense

Black -> Ground Wires

Red -> 3V3 Wires

Green -> Connective Wires

Figure 11: Fritzing Circuit Schematic for CarpalSense



Fusion Models

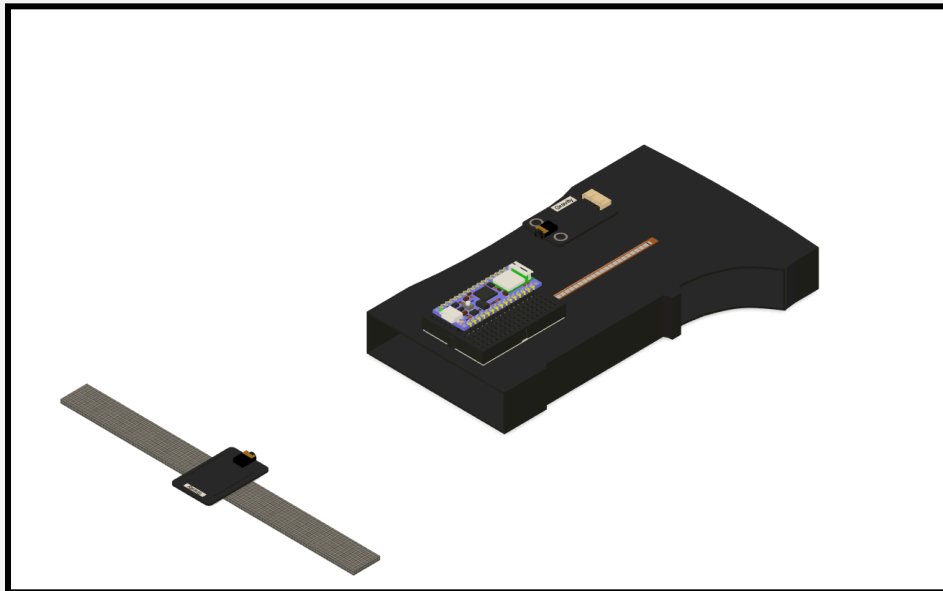
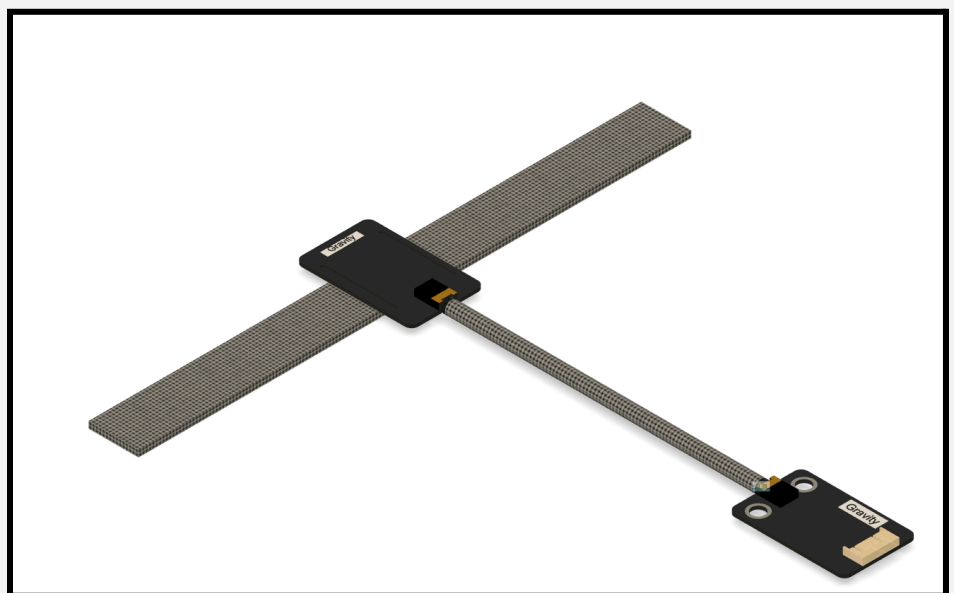


Figure 12:
Fusion 360
Model of
CarpalSense
Features
Arduino Nano,
Flex Sensor and
EMG

Figure 13: Fusion
360 Custom Model of
Oymotion EMG
Sensor

Top Left: EMG Side of
sensor attached to strap
Top Right: Receiver Side to
connect to Arduino



Data

Flex

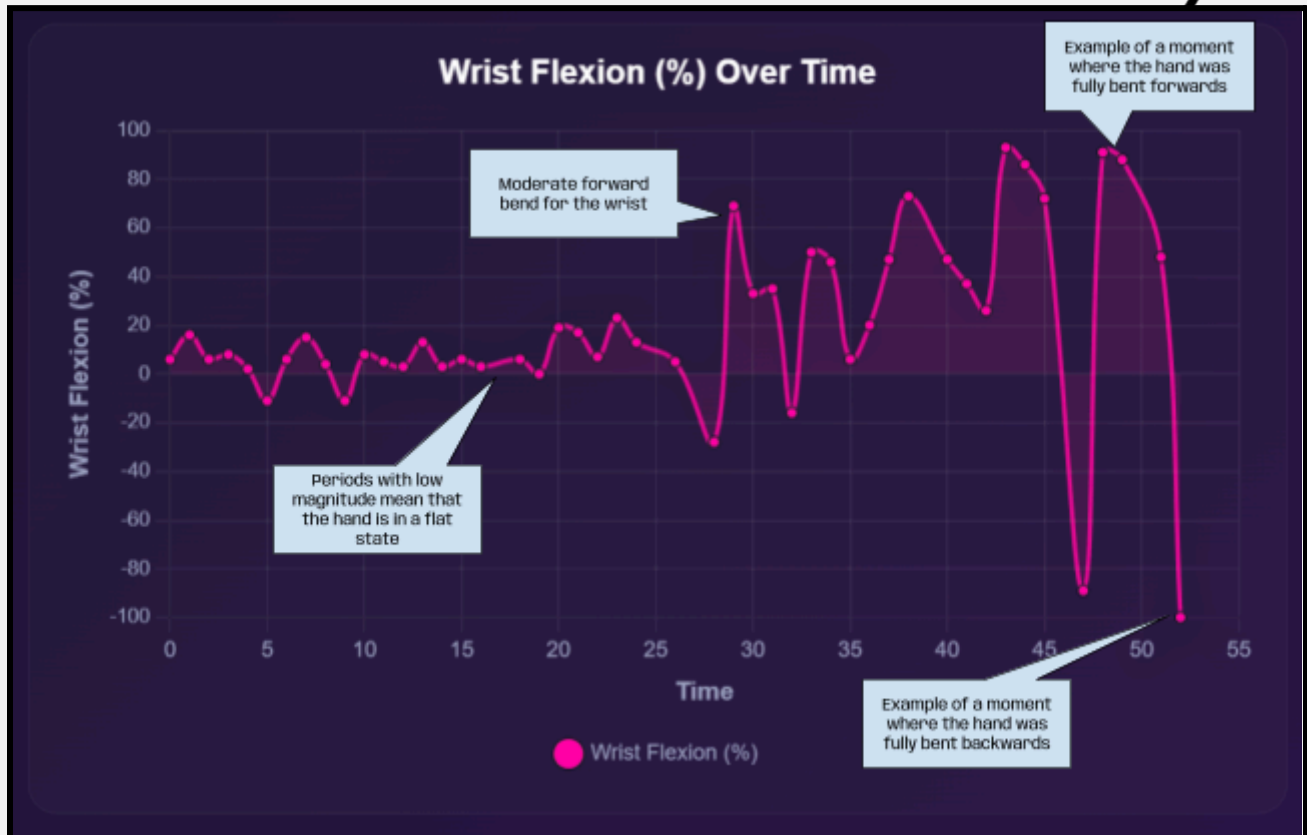


Figure 14: Example Graph of Data Recorded by Flex Sensor with notes detailing graph specifics

Positive Percentages = Forward Bend
Negative Percentages = Backward Bend

Data

EMG

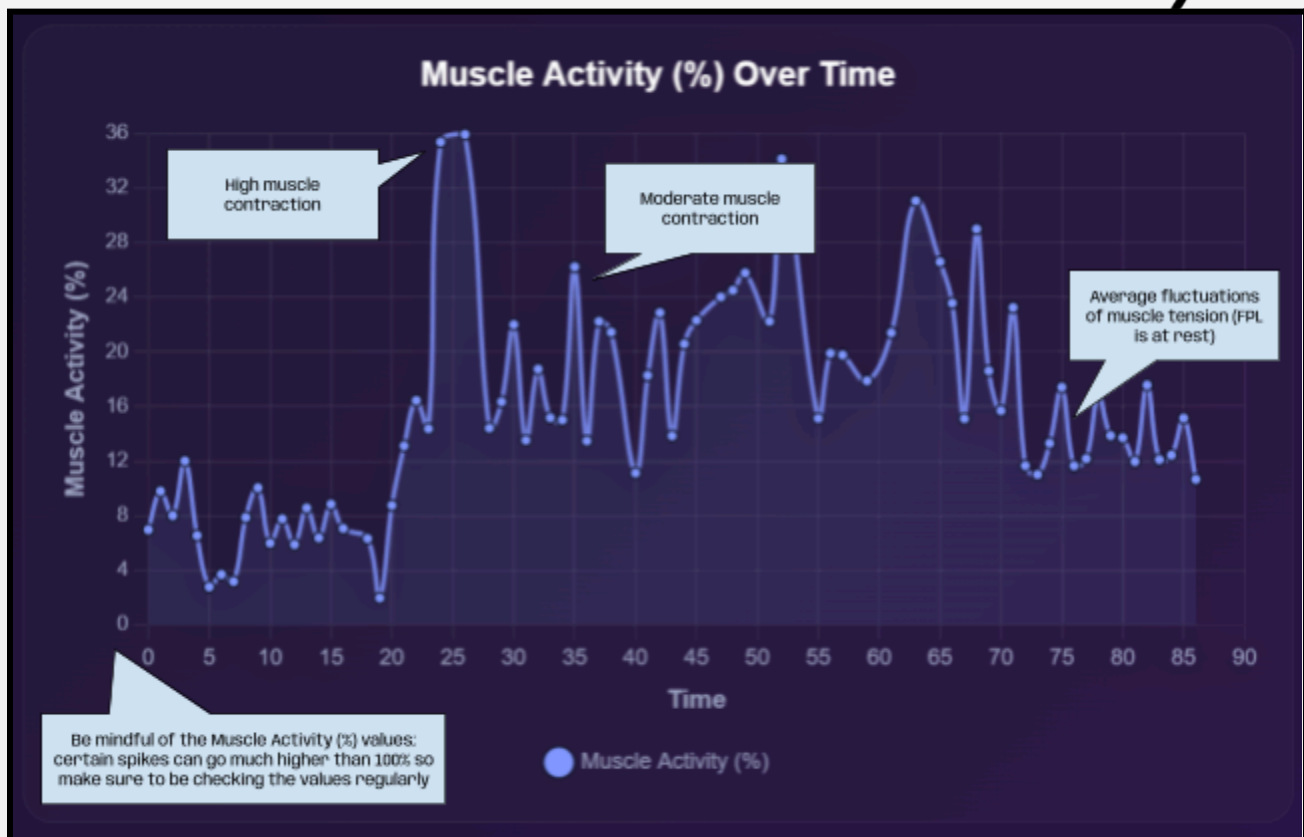


Figure 15: Example Graph of Data Recorded by Dry EMG Sensor with notes detailing graph specifics

Higher Percentage = Stronger Contraction

A percentage >100% does **not** mean the muscle is being overstrained

Troubleshooting

Flex

❖ Bending not recording

- Double check all wires and make sure they are connected
 - Wires will connect to the Clinch Connector inside of the housing, not the flex sensor itself
- Check the flex sensor's position
 - The end of the sensor should be pushed as far in as it can go.
 - If the sensor is not fully pushed in, gently push it further in at the position specified by the green circle on **Figure 15**.



Figure 16: Close-up of the sensor shows where to push back as well as ideal location of the end of the sensor

Troubleshooting

EMG

❖ Contractions not recording

- Check to see if EMG sensor is placed on the right muscle body
 - Place your arm palm up and clench your fist repeatedly
 - Feel along the side of your forearm until you find a muscle that is tensing with your forearm. Place EMG there.
- Make sure all wires are connected properly to the sensors
- Tighten EMG sensor on the muscle body

❖ Constantly Returning “High” Contractions

- Restart the program
 - Unplug the sensor and plug it back in
 - This will recalibrate the baseline
- Double check EMG placement on the forearm
 - Incorrect placement of the EMG can cause it to read multiple muscles at once or pick up on unnecessary noise
 - View placement directions above

❖ General Debugging

- Bicep Check
 - If there are concerns with the EMG sensor, place the sensor on your bicep and monitor the contractions following the same procedure as before
 - If the contractions align with bicep motions, it is most likely a placement error
 - If the contractions do not align, check all wires connections to make sure the EMG is properly connected



Figure 17: Full forearm view of the EMG sensor. Details where to place sensor